

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:02:08

Source: data/json/noise_experiments/v2/noise2i_anon_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_anon/noise2i_anon_game_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_anon
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_1=\$7, Agent_2=\$4;

noise/visible: sent visible=\$3.63, received visible=\$10.89, returned visible=\$4.77

Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.0,

"Agent_2": 4.0

}

Visible balances after round: {

"Agent_1": 6.77,

"Agent_2": 5.8900000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as the SENDER, I need to balance trust-building with risk management.

Since:

- Sending money creates value (3x multiplier)
- I have no history with this receiver yet
- There's communication noise that might affect perceptions

I'll send a moderate amount that shows willingness to cooperate while managing risk.

```
{'send': 3}
```

This gives the receiver \$9 to work with, and if they return a fair portion, we both benefit. If they return at least \$4, I'll end up better than my initial \$5.

Reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game problem

Key considerations:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- There's uncertainty about whether the receiver will reciprocate
- This is round 1, so I have no history with this receiver
- There's mentioned "communication noise" which means amounts may differ from what's actually sent

Strategic thinking:

- Sending some amount creates mutual benefit opportunities
- Sending everything (\$5) shows maximum trust but is risky
- Sending a moderate amount (like \$2-3) balances trust-building with risk management
- Since this is round 1 and I don't know the receiver's behavior, a moderate approach seems reasonable

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to trust and create value
- Leaves me with \$2 as a safety net
- Gives the receiver \$9 to work with
- If they return even \$4, I'd end up with \$6 (better than my initial \$5)

This seems like a balanced approach for the first round.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:58:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
}
```

```
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to The sender? (0-10.89)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- This gave me \$10.89 to work with
- The sender currently has only \$1.37 remaining

To reciprocate their trust and ensure mutual benefit:

- If I return \$5.00, the sender would end up with \$6.37 total (a gain of \$1.37)
- I would keep \$5.89 (also a good outcome)

This seems fair given the sender took the initial risk and showed high trust.

{'return': 5.00}

Reasoning:

Let me understand this situation:

1. The sender received \$5 (their endowment)
2. The sender sent \$3.63 (72.6% of \$5)
3. This amount was multiplied by 3x: $\$3.63 \times 3 = \10.89
4. I received \$10.89
5. I need to decide how much to return (between \$0 and \$10.89)

The sender showed significant trust by sending 72.6% of their endowment. Let me think about what would be fair:

- The sender currently has: $\$5 - \$3.63 = \$1.37$
- I currently have: $\$10.89$
- If I return nothing, sender gets $\$1.37$ total, I get $\$10.89$
- If I return everything, sender gets $\$1.37 + \$10.89 = \$12.26$, I get $\$0$

A fair approach might be to:

1. Split the total gain equally, OR
2. Return enough so the sender ends up with more than they started with

Total in the system: $\$1.37$ (sender's remaining) + $\$10.89$ (what I have) = $\$12.26$
This represents a gain of $\$12.26 - \$5 = \$7.26$ from the initial $\$5$

If we split equally:

- Each person should get: $\$12.26 / 2 = \6.13
- Sender currently has $\$1.37$, so I should return: $\$6.13 - \$1.37 = \$4.76$
- I would keep: $\$10.89 - \$4.76 = \$6.13$

Alternatively, I could think about returning a similar proportion to what they sent. They sent 72.6% of their endowment. If I return around 40-50% of what I received, that would be:

- 40% of $\$10.89 = \4.36
- 50% of $\$10.89 = \5.45

I could also ensure the sender ends up better off than their starting position. Returning $\$4$ gives them $\$5.37$, returning $\$5$ gives them $\$6.37$ (a 28% gain), or returning $\$6$ gives them $\$7.37$ (47% gain). Given their substantial trust in sending most of their endowment, I should return a meaningful amount-somewhere in the $\$4.50$ - $\$5.50$ range seems appropriate to reciprocate their generosity while still benefiting from the exchange....

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 1014,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent $\$3.5$; received $\$10.5$; Agent_1 returned $\$0$; payoffs Agent_2= $\$1.5$, Agent_1= $\$10.5$; noise/visible: sent visible= $\$0$, received visible= $\$0$
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 10.5, "min": 0, "raw": 0.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 17.5,
"Agent_2": 5.5

}

Visible balances after round: {

"Agent_1": 6.77,
"Agent_2": 7.3900000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",

```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to The sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- This gave me \$10.89 to work with
- The sender currently has only \$1.37 remaining

To reciprocate their trust and ensure mutual benefit:

- If I return \$5.00, the sender would end up with \$6.37 total (a gain of \$1.37)
- I would keep \$5.89 (also a good outcome)

This seems fair given the sender took the initial risk and showed high trust.

reasoning:

1. The sender received \$5 (their endowment)
2. The sender sent \$3.63 (72.6% of \$5)
3. This amount was multiplied by 3x: $\$3.63 \times 3 = \10.89
4. I received \$10.89
5. I need to decide how much to return (between \$0 and \$10.89)

- The sender currently has: $\$5 - \$3.63 = \$1.37$
- I currently have: $\$10.89$
- If I return nothing, sender gets $\$1.37$ total, I get $\$10.89$
- If I return everything, sender gets $\$1.37 + \$10.89 = \$12.26$, I get $\$0$

1. Split the total gain equally, OR
2. Return enough so the sender ends up with more than they started with

- Each person should get: $\$12.26 / 2 = \6.13
- Sender currently has \$1.37, so I should return: $\$6.13 - \$1.37 = \$4.76$
- I would keep: $\$10.89 - \$4.76 = \$6.13$

- 40% of \$10.89 = \$4.36
- 50% of \$10.89 = \$5.45

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1014,
  "reasoning_tokens": 0
}
```

content:

1.

Assistant output:

```
{'send': 3.50}
```

- I'm now the sender in Round 2
- I have \$5 to potentially send
- My total earnings so far are \$5.89 (from Round 1)

Page 8 of 65

Now I need to decide how much to send. Let me think about this:

1. The previous sender showed high trust (72.6%)
2. This could be the same person or a different person - I don't know
3. If I want to maintain reciprocity and encourage cooperation, I should also show trust
4. Sending around 70% would be consistent with what I received: 70% of \$5 = \$3.50

Let me calculate outcomes:

- If I send \$3.50:
 - Receiver gets \$10.50
 - If they return half (~\$5.25), I'd end with \$6.75 total (\$1.50 + \$5.25)
 - That would be a gain of \$1.75
- If I send \$4.00:
 - Receiver gets \$12.00
 - If they return half (\$6.00), I'd end with \$7.00 total (\$1.00 + \$6.00)
 - That would be a gain of \$2.00

Given that the previous sender showed high trust (72.6%), I should reciprocate with similar trust. I'll send \$3.50, which is 70% of my endowment, similar to what was sent to me.

Actually, let me be more precise - the sender sent 72.6%, which was \$3.63. So 72.6% of \$5 is \$3.63.

I'll send \$3.50 or \$3.63 to show similar trust levels. Let me go with \$3.50 as a nice round number that shows high trust.

```
Usage: {
  "input_tokens": 585,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the SENDER, I need to balance trust-building with risk management.

Since:

- Sending money creates value (3x multiplier)
- I have no history with this receiver yet
- There's communication noise that might affect perceptions

I'll send a moderate amount that shows willingness to cooperate while managing risk.

{'send': 3}

This gives the receiver \$9 to work with, and if they return a fair portion, we both benefit. If they return at least \$4, I'll end up better than my initial \$5.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game problem

Key considerations:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- There's uncertainty about whether the receiver will reciprocate
- This is round 1, so I have no history with this receiver
- There's mentioned "communication noise" which means amounts may differ from what's actually sent

Strategic thinking:

- Sending some amount creates mutual benefit opportunities
- Sending everything (\$5) shows maximum trust but is risky
- Sending a moderate amount (like \$2-3) balances trust-building with risk management
- Since this is round 1 and I don't know the receiver's behavior, a moderate approach seems reasonable

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to trust and create value
- Leaves me with \$2 as a safety net
- Gives the receiver \$9 to work with
- If they return even \$4, I'd end up with \$6 (better than my initial \$5)

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.77.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier: $0 \times 3 = \$0$).

I have no money to return in this round.

```
{'return': 0}
```

Reasoning:

Let me understand this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$6.77
3. The sender sent me \$0, so I received \$0 (which gets multiplied by 3, but $0 \times 3 = 0$)
4. I need to decide how much to return (from 0 to 0)

Wait, let me think about my earnings:

- Round 1: I was SENDER and sent \$3
- So I kept \$5 - \$3 = \$2
- The receiver got $\$3 \times 3 = \9
- If my total earnings are \$6.77, that means I must have received $\$6.77 - \$2 = \$4.77$ back in Round 1

That's interesting - the receiver returned \$4.77 out of the \$9 they received, which is about 53%. That's actually quite generous.

Now in Round 2:

- The sender sent me \$0
- So I received \$0 ($0 \times 3 = \0)
- I can return between \$0 and \$0

Since I received nothing, I have nothing to return. The only possible response is to return \$0

Also, there's the note about communication noise - maybe the sender intended to send something but due to noise it appears as \$0? Or maybe they actually sent \$0, possibly as retaliation for something or just being cautious.

Either way, I can only return what I have, which is \$0.

Usage: {

```
"input_tokens": 553,
"output_tokens": 409,
"reasoning_tokens": 0
}
```

=====

ROUND 3

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

```

dyad_1: Agent_1 sent $3; received $9; Agent_2 returned $0; payoffs Agent_1=$2, Agent_2=$9;
noise/visible: sent visible=$0, received visible=$0, returned visible=$0.29
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw":
    0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 19.5,
  "Agent_2": 14.5
}
Visible balances after round: {
  "Agent_1": 9.059999999999999,
  "Agent_2": 7.390000000000001
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the SENDER, I need to balance trust-building with risk management.

Since:

- Sending money creates value (3x multiplier)
- I have no history with this receiver yet
- There's communication noise that might affect perceptions

I'll send a moderate amount that shows willingness to cooperate while managing risk.

```
{'send': 3}
```

This gives the receiver \$9 to work with, and if they return a fair portion, we both benefit. If they return at least \$4, I'll end up better than my initial \$5.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game problem

Key considerations:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- There's uncertainty about whether the receiver will reciprocate
- This is round 1, so I have no history with this receiver
- There's mentioned "communication noise" which means amounts may differ from what's actually sent

Strategic thinking:

- Sending some amount creates mutual benefit opportunities
- Sending everything (\$5) shows maximum trust but is risky
- Sending a moderate amount (like \$2-3) balances trust-building with risk management
- Since this is round 1 and I don't know the receiver's behavior, a moderate approach seems reasonable

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to trust and create value
- Leaves me with \$2 as a safety net
- Gives the receiver \$9 to work with
- If they return even \$4, I'd end up with \$6 (better than my initial \$5)

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.77.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier: $0 \times 3 = \$0$).

I have no money to return in this round.

`{'return': 0}`

reasoning:

Let me understand this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$6.77
3. The sender sent me \$0, so I received \$0 (which gets multiplied by 3, but $0 \times 3 = 0$)
4. I need to decide how much to return (from 0 to 0)

Wait, let me think about my earnings:

- Round 1: I was SENDER and sent \$3
- So I kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- If my total earnings are \$6.77, that means I must have received $\$6.77 - \$2 = \$4.77$ back in Round 1

That's interesting - the receiver returned \$4.77 out of the \$9 they received, which is about 53%. That's actually quite generous.

Now in Round 2:

- The sender sent me \$0
- So I received \$0 ($0 \times 3 = \0)
- I can return between \$0 and \$0

Since I received nothing, I have nothing to return. The only possible response is to return \$0.

Also, there's the note about communication noise - maybe the sender intended to send something but due to noise it appears as \$0? Or maybe they actually sent \$0, possibly as retaliation for something or just being cautious.

Either way, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 553,
  "output_tokens": 409,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.77.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my history:

Round 1: I sent \$3 and my current total is \$6.77, suggesting the receiver returned approximately \$4.77 - a cooperative response.

Round 2: The sender sent me \$0, which was uncooperative.

For Round 3, I'm back as SENDER with a new (or possibly the same) receiver. The good return in Round 1 shows that cooperation can be rewarding. I'll continue with a cooperative strategy.

`{'send': 3}`

This maintains consistency, signals trustworthiness, and creates value through the 3x multiplier. If this receiver is similarly cooperative, we'll both benefit.

Reasoning:

Let me analyze my situation:

Round 1: I was the SENDER and sent \$3

- I kept: $\$5 - \$3 = \$2$
- Receiver got: $\$3 \times 3 = \9
- My total earnings are now \$6.77

So the receiver must have returned: $\$6.77 - \$2 = \$4.77$

Round 2: I was the RECEIVER and the sender sent me \$0

- I received \$0, returned \$0
- No change to my earnings from this round

So my earnings of \$6.77 come from Round 1 where I sent \$3 and received back \$4.77.

Wait, but there's communication noise mentioned. Let me reconsider.

Actually, if the receiver returned \$4.77, that's pretty generous - they kept $\$9 - \$4.77 = \$4.23$ and gave me back \$4.77. That's more than half!

Now in Round 3, I'm playing with a different person (or maybe the same person? It's not clear)

. But I should note:

- Round 1 receiver was cooperative (returned a good amount)
- Round 2 sender sent nothing

For Round 3, I'm the sender again. This might be a different receiver than in Round 1. Given:

1. The good experience in Round 1 suggests cooperation can work
2. The poor experience in Round 2 (though I was receiver) shows not everyone cooperates
3. I should continue trying to build cooperative relationships

I think I'll send a similar amount to Round 1, maybe slightly adjusted. Sending \$3 worked well before. I could try \$3 again, or perhaps \$2 to be slightly more cautious given the Round 2 experience.

Let me send \$3 again to signal I'm still willing to cooperate.

```
Usage: {  
  "input_tokens": 652,  
  "output_tokens": 575,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to The sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- This gave me \$10.89 to work with
- The sender currently has only \$1.37 remaining

To reciprocate their trust and ensure mutual benefit:

- If I return \$5.00, the sender would end up with \$6.37 total (a gain of \$1.37)
- I would keep \$5.89 (also a good outcome)

This seems fair given the sender took the initial risk and showed high trust.

{'return': 5.00}

reasoning:

Let me understand this situation:

1. The sender received \$5 (their endowment)
2. The sender sent \$3.63 (72.6% of \$5)
3. This amount was multiplied by 3x: $\$3.63 \times 3 = \10.89
4. I received \$10.89
5. I need to decide how much to return (between \$0 and \$10.89)

The sender showed significant trust by sending 72.6% of their endowment. Let me think about what would be fair:

- The sender currently has: $\$5 - \$3.63 = \$1.37$
- I currently have: \$10.89
- If I return nothing, sender gets \$1.37 total, I get \$10.89
- If I return everything, sender gets $\$1.37 + \$10.89 = \$12.26$, I get \$0

A fair approach might be to:

1. Split the total gain equally, OR
2. Return enough so the sender ends up with more than they started with

Total in the system: \$1.37 (sender's remaining) + \$10.89 (what I have) = \$12.26
This represents a gain of \$12.26 - \$5 = \$7.26 from the initial \$5

If we split equally:

- Each person should get: $\$12.26 / 2 = \6.13
- Sender currently has \$1.37, so I should return: $\$6.13 - \$1.37 = \$4.76$
- I would keep: $\$10.89 - \$4.76 = \$6.13$

Alternatively, I could think about returning a similar proportion to what they sent. They sent 72.6% of their endowment. If I return around 40-50% of what I received, that would be:

- 40% of \$10.89 = \$4.36
- 50% of \$10.89 = \$5.45

I could also ensure the sender ends up better off than their starting position. Returning \$4 gives them \$5.37, returning \$5 gives them \$6.37 (a 28% gain), or returning \$6 gives them \$7.37 (47% gain). Given their substantial trust in sending most of their endowment, I should return a meaningful amount-somewhere in the \$4.50-\$5.50 range seems appropriate to reciprocate their generosity while still benefiting from the exchange....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1014,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.89000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed high trust by sending 72.6% of their endowment, and I reciprocated fairly.

I want to maintain this cooperative approach. I'll send a similar proportion to demonstrate trust and encourage reciprocity.

```
{'send': 3.50}
```

reasoning:

Looking at my situation:

- I'm now the sender in Round 2
- I have \$5 to potentially send
- My total earnings so far are \$5.89 (from Round 1)

In Round 1, the sender sent me \$3.63 (72.6% of their endowment), showing high trust. I returned \$5.00, which was a fair reciprocation.

Now I need to decide how much to send. Let me think about this:

1. The previous sender showed high trust (72.6%)
2. This could be the same person or a different person - I don't know
3. If I want to maintain reciprocity and encourage cooperation, I should also show trust
4. Sending around 70% would be consistent with what I received: 70% of \$5 = \$3.50

Let me calculate outcomes:

- If I send \$3.50:
 - Receiver gets \$10.50
 - If they return half (~\$5.25), I'd end with \$6.75 total (\$1.50 + \$5.25)
 - That would be a gain of \$1.75
- If I send \$4.00:
 - Receiver gets \$12.00
 - If they return half (\$6.00), I'd end with \$7.00 total (\$1.00 + \$6.00)
 - That would be a gain of \$2.00

Given that the previous sender showed high trust (72.6%), I should reciprocate with similar trust. I'll send \$3.50, which is 70% of my endowment, similar to what was sent to me.

Actually, let me be more precise - the sender sent 72.6%, which was \$3.63. So 72.6% of \$5 is \$3.63.

I'll send \$3.50 or \$3.63 to show similar trust levels. Let me go with \$3.50 as a nice rounded number that shows high trust.

usage:

```
{
  "input_tokens": 585,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.390000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, so I received \$0. This means I have nothing to return - the maximum I can return is \$0.

This is interesting - in round 2, I sent \$3.50 (70% of my endowment) showing trust, but the sender this round sent me nothing. This could be:

1. A different sender in this round (multi-player game)
2. Communication noise caused them to receive/perceive something different
3. They chose not to send anything

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

Usage: {

```
"input_tokens": 715,
"output_tokens": 171,
"reasoning_tokens": 0
}
```

=====

ROUND 4

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2.5; received \$7.5; Agent_1 returned \$1; payoffs Agent_2=\$3.5, Agent_1=\$6.5; noise/visible: sent visible=\$0.68, received visible=\$2.04, returned visible=\$1.98

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 7.5, "min": 0, "raw": 1.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

```
"Agent_1": 26.0,
"Agent_2": 18.0
}
```

Visible balances after round: {

```
"Agent_1": 10.099999999999998,
"Agent_2": 11.870000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:55

Temperature: 0.8

Exact input messages sent to model:

content:

GAME :

When you are the SENDER:

- When you are the RECEIVER:

- MYTH WRITING:

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.89000000000000

1.

How much do you send to the receiver this round? (0-5)

Page 19 of 65

content:

Looking at the previous round, the sender showed high trust by sending 72.6% of their endowment, and I reciprocated fairly.

I want to maintain this cooperative approach. I'll send a similar proportion to demonstrate trust and encourage reciprocity.

```
{'send': 3.50}
```

reasoning:

Looking at my situation:

- I'm now the sender in Round 2
- I have \$5 to potentially send
- My total earnings so far are \$5.89 (from Round 1)

In Round 1, the sender sent me \$3.63 (72.6% of their endowment), showing high trust. I returned \$5.00, which was a fair reciprocation.

Now I need to decide how much to send. Let me think about this:

1. The previous sender showed high trust (72.6%)
2. This could be the same person or a different person - I don't know
3. If I want to maintain reciprocity and encourage cooperation, I should also show trust
4. Sending around 70% would be consistent with what I received: 70% of \$5 = \$3.50

Let me calculate outcomes:

- If I send \$3.50:
 - Receiver gets \$10.50
 - If they return half (~\$5.25), I'd end with \$6.75 total (\$1.50 + \$5.25)
 - That would be a gain of \$1.75
- If I send \$4.00:
 - Receiver gets \$12.00
 - If they return half (\$6.00), I'd end with \$7.00 total (\$1.00 + \$6.00)
 - That would be a gain of \$2.00

Given that the previous sender showed high trust (72.6%), I should reciprocate with similar trust. I'll send \$3.50, which is 70% of my endowment, similar to what was sent to me.

Actually, let me be more precise - the sender sent 72.6%, which was \$3.63. So 72.6% of \$5 is \$3.63.

I'll send \$3.50 or \$3.63 to show similar trust levels. Let me go with \$3.50 as a nice round number that shows high trust.

usage:

```
{
  "input_tokens": 585,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.390000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant**content:**

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0. This means I have nothing to return - the maximum I can return is \$0.

This is interesting - in round 2, I sent \$3.50 (70% of my endowment) showing trust, but the sender this round sent me nothing. This could be:

1. A different sender in this round (multi-player game)
2. Communication noise caused them to receive/perceive something different

Since I received \$0, I can only return \$0.

Message 6 role=user
content:
Round 4

How much do you send to the receiver this round? (0-5)

Looking at the pattern:

- Round 2: I sent \$3.50 and received \$0 back
- Round 3: I received \$0 from the sender

I'll send a moderate amount to signal willingness to cooperate while being somewhat more cautious:

Let me think about what happened in the previous rounds:

- Started Round 2 with: \$5.89 (from Round 1)
- Received in Round 2: \$5 (endowment)
- Sent: \$3.50
- So I had: $\$5 - \$3.50 = \$1.50$ left
- Then I received back some amount from the receiver
- Total after Round 2 = Previous + \$1.50 + returned = $\$5.89 + \$1.50 + \text{returned} = \7.39
- So returned = $\$7.39 - \$7.39 = \$0$

- Before Round 2, I had: \$5.89
- In Round 2, I got \$5 endowment, sent \$3.50, so kept \$1.50
- Then receiver returned something to me
- After Round 2, total = \$7.39
- So: \$5.89 + \$1.50 + returned = \$7.39
- returned = \$7.39 - \$5.89 = \$1.50

Now it's Round 4 and I'm sending again. Looking back, the receiver didn't return anything after I sent \$3.50 in Round 2, and then the sender sent me nothing in Round 3. There's communication noise that could distort amounts, so it's possible the receiver did send something back but it appeared as \$0, or the sender in Round 3 actually sent something that got lost. I need to decide whether to trust again or protect myself given this history of apparent defection.

Page 21 of 65

```
    exposing myself to another potential loss....
Usage: {
  "input_tokens": 591,
  "output_tokens": 1069,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.77.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 (even after the 3x multiplier: $0 \times 3 = \$0$).

I have no money to return in this round.

{'return': 0}

reasoning:

Let me understand this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$6.77
3. The sender sent me \$0, so I received \$0 (which gets multiplied by 3, but $0 \times 3 = 0$)
4. I need to decide how much to return (from 0 to 0)

Wait, let me think about my earnings:

- Round 1: I was SENDER and sent \$3
- So I kept $\$5 - \$3 = \$2$
- The receiver got $\$3 \times 3 = \9
- If my total earnings are \$6.77, that means I must have received $\$6.77 - \$2 = \$4.77$ back in Round 1

That's interesting - the receiver returned \$4.77 out of the \$9 they received, which is about 53%. That's actually quite generous.

Now in Round 2:

- The sender sent me \$0
- So I received \$0 ($0 \times 3 = \0)
- I can return between \$0 and \$0

Since I received nothing, I have nothing to return. The only possible response is to return \$0.

Also, there's the note about communication noise - maybe the sender intended to send something but due to noise it appears as \$0? Or maybe they actually sent \$0, possibly as retaliation for something or just being cautious.

Either way, I can only return what I have, which is \$0.

usage:

```
{
  "input_tokens": 553,
  "output_tokens": 409,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.77.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my history:

Round 1: I sent \$3 and my current total is \$6.77, suggesting the receiver returned approximately \$4.77 - a cooperative response.

Round 2: The sender sent me \$0, which was uncooperative.

For Round 3, I'm back as SENDER with a new (or possibly the same) receiver. The good return in Round 1 shows that cooperation can be rewarding. I'll continue with a cooperative strategy.

```
{'send': 3}
```

This maintains consistency, signals trustworthiness, and creates value through the 3x multiplier. If this receiver is similarly cooperative, we'll both benefit.

reasoning:

Let me analyze my situation:

Round 1: I was the SENDER and sent \$3

- I kept: $\$5 - \$3 = \$2$
- Receiver got: $\$3 \times 3 = \9
- My total earnings are now \$6.77

So the receiver must have returned: $\$6.77 - \$2 = \$4.77$

Round 2: I was the RECEIVER and the sender sent me \$0

- I received \$0, returned \$0
- No change to my earnings from this round

So my earnings of \$6.77 come from Round 1 where I sent \$3 and received back \$4.77.

Wait, but there's communication noise mentioned. Let me reconsider.

Actually, if the receiver returned \$4.77, that's pretty generous - they kept $\$9 - \$4.77 = \$4.23$ and gave me back \$4.77. That's more than half!

Now in Round 3, I'm playing with a different person (or maybe the same person? It's not clear). But I should note:

- Round 1 receiver was cooperative (returned a good amount)
- Round 2 sender sent nothing

For Round 3, I'm the sender again. This might be a different receiver than in Round 1. Given:

1. The good experience in Round 1 suggests cooperation can work
2. The poor experience in Round 2 (though I was receiver) shows not everyone cooperates
3. I should continue trying to build cooperative relationships

I think I'll send a similar amount to Round 1, maybe slightly adjusted. Sending \$3 worked well before. I could try \$3 again, or perhaps \$2 to be slightly more cautious given the Round 2 experience.

Let me send \$3 again to signal I'm still willing to cooperate.

usage:

```
{
  "input_tokens": 652,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.059999999999999.

This round, The sender sent you \$0.68, so you received \$2.04. How much do you return to the sender? (0-2.04)

Assistant output:

As RECEIVER this round:

- The sender sent \$0.68
- I received \$2.04 (after 3x multiplier)
- The sender showed some cooperation by sending a positive amount

To maintain reciprocity and encourage continued cooperation, I'll return a fair portion. Since the sender did cooperate (even if with a smaller amount), I should reward that trust.

```
{'return': 1.0}
```

This gives the sender a positive return on their investment ($\$5 - \$0.68 + \$1.00 = \5.32) while I keep \$1.04 from this round.

Reasoning:

Let me analyze this situation:

As RECEIVER this round:

- The sender sent \$0.68
- I received $\$0.68 \times 3 = \2.04
- I need to decide how much to return (0-2.04)

My total earnings so far are \$9.06.

Looking at the pattern:

- Round 1: I was sender, sent \$3, got a good return (around \$4.77)
- Round 2: I was receiver, received \$0, returned \$0
- Round 3: I was sender, sent \$3
- Round 4: I am receiver, received \$2.04

The sender in Round 4 sent \$0.68, which is about 13.6% of their \$5 endowment. This is a relatively small amount compared to my \$3 sends, but it's still cooperative (non-zero).

Given that:

1. The note says there may be communication noise
2. The sender did cooperate by sending something
3. I've been cooperative in the past
4. Fair reciprocity would be to return a reasonable amount

A fair return might be:

- If I return about half (\$1.02), the sender would get: $\$5 - \$0.68 + \$1.02 = \5.34
- I would keep: $\$2.04 - \$1.02 = \$1.02$

Or I could be more generous:

- If I return \$1.36 (2/3), the sender would get: $\$5 - \$0.68 + \$1.36 = \5.68
- I would keep: $\$2.04 - \$1.36 = \$0.68$

My strategy has been cooperative. Given the sender sent a smaller amount, I should still return a fair portion. Returning about half to 2/3 seems reasonable. Let me return \$1.00 which is roughly half.

Actually, let me think about this more carefully. The sender sent \$0.68. After 3x, I got \$2.04

.
...

```
Usage: {  
  "input_tokens": 711,  
  "output_tokens": 692,  
  "reasoning_tokens": 0  
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$0; payoffs Agent_1=\$2, Agent_2=\$9;
noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.77
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 28.0,
"Agent_2": 27.0

}

Visible balances after round: {

"Agent_1": 12.869999999999997,
"Agent_2": 11.870000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.77.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my history:

Round 1: I sent \$3 and my current total is \$6.77, suggesting the receiver returned approximately \$4.77 - a cooperative response.

Round 2: The sender sent me \$0, which was uncooperative.

For Round 3, I'm back as SENDER with a new (or possibly the same) receiver. The good return in Round 1 shows that cooperation can be rewarding. I'll continue with a cooperative strategy.

{'send': 3}

This maintains consistency, signals trustworthiness, and creates value through the 3x multiplier. If this receiver is similarly cooperative, we'll both benefit.

reasoning:

Let me analyze my situation:

Round 1: I was the SENDER and sent \$3

- I kept: $\$5 - \$3 = \$2$
- Receiver got: $\$3 \times 3 = \9
- My total earnings are now \$6.77

So the receiver must have returned: $\$6.77 - \$2 = \$4.77$

Round 2: I was the RECEIVER and the sender sent me \$0

- I received \$0, returned \$0
- No change to my earnings from this round

So my earnings of \$6.77 come from Round 1 where I sent \$3 and received back \$4.77.

Wait, but there's communication noise mentioned. Let me reconsider.

Actually, if the receiver returned \$4.77, that's pretty generous - they kept $\$9 - \$4.77 = \$4.23$ and gave me back \$4.77. That's more than half!

Now in Round 3, I'm playing with a different person (or maybe the same person? It's not clear). But I should note:

- Round 1 receiver was cooperative (returned a good amount)
- Round 2 sender sent nothing

For Round 3, I'm the sender again. This might be a different receiver than in Round 1. Given:

1. The good experience in Round 1 suggests cooperation can work
2. The poor experience in Round 2 (though I was receiver) shows not everyone cooperates
3. I should continue trying to build cooperative relationships

I think I'll send a similar amount to Round 1, maybe slightly adjusted. Sending \$3 worked well before. I could try \$3 again, or perhaps \$2 to be slightly more cautious given the Round 2 experience.

Let me send \$3 again to signal I'm still willing to cooperate.

usage:

```
{
  "input_tokens": 652,
  "output_tokens": 575,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.059999999999999.

This round, The sender sent you \$0.68, so you received \$2.04. How much do you return to the sender? (0-2.04)

Message 5 role=assistant

content:

As RECEIVER this round:

- The sender sent \$0.68
- I received \$2.04 (after 3x multiplier)

- The sender showed some cooperation by sending a positive amount

To maintain reciprocity and encourage continued cooperation, I'll return a fair portion. Since the sender did cooperate (even if with a smaller amount), I should reward that trust.

```
{'return': 1.0}
```

This gives the sender a positive return on their investment ($\$5 - \$0.68 + \$1.00 = \5.32) while I keep \$1.04 from this round.

reasoning:

Let me analyze this situation:

As RECEIVER this round:

- The sender sent \$0.68
- I received $\$0.68 \times 3 = \2.04
- I need to decide how much to return (0-2.04)

My total earnings so far are \$9.06.

Looking at the pattern:

- Round 1: I was sender, sent \$3, got a good return (around \$4.77)
- Round 2: I was receiver, received \$0, returned \$0
- Round 3: I was sender, sent \$3
- Round 4: I am receiver, received \$2.04

The sender in Round 4 sent \$0.68, which is about 13.6% of their \$5 endowment. This is a relatively small amount compared to my \$3 sends, but it's still cooperative (non-zero).

Given that:

1. The note says there may be communication noise
2. The sender did cooperate by sending something
3. I've been cooperative in the past
4. Fair reciprocity would be to return a reasonable amount

A fair return might be:

- If I return about half (\$1.02), the sender would get: $\$5 - \$0.68 + \$1.02 = \5.34
- I would keep: $\$2.04 - \$1.02 = \$1.02$

Or I could be more generous:

- If I return \$1.36 (2/3), the sender would get: $\$5 - \$0.68 + \$1.36 = \5.68
- I would keep: $\$2.04 - \$1.36 = \$0.68$

My strategy has been cooperative. Given the sender sent a smaller amount, I should still return a fair portion. Returning about half to 2/3 seems reasonable. Let me return \$1.00 which is roughly half.

Actually, let me think about this more carefully. The sender sent \$0.68. After 3x, I got \$2.04.

...

usage:

```
{
  "input_tokens": 711,
  "output_tokens": 692,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.099999999999999.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my trajectory:

- Round 3: I sent \$3 (cooperative approach)
- Round 4: As receiver, sender sent \$0.68 (some cooperation), I returned \$1.00 to reciprocate

Total earnings increased from \$9.06 to \$10.10, showing positive outcomes from cooperation. The pattern shows that cooperative play has been generally rewarding, though there's variation

in partners' cooperation levels.

For Round 5, I'll maintain a cooperative strategy:

{'send': 3}

This amount:

- Creates value through the 3x multiplier (\$9 for receiver)
- Shows continued trustworthiness
- Has historically led to positive returns
- Balances risk with potential reward

Reasoning:

Let me track my earnings history:

Round 1: I sent \$3, total earnings were \$6.77

Round 2: I was receiver, sender sent me \$0

Round 3: I sent \$3, total increased to \$9.06

Round 4: I was receiver, sender sent \$0.68, I received \$2.04, I returned \$1.00, total is now \$10.10

Let me verify Round 4:

- Started with \$9.06
- Received \$2.04
- Returned \$1.00
- New total: $\$9.06 + \$2.04 - \$1.00 = \10.10 ?

Now looking at the pattern:

- Round 1: I sent \$3, got back good return (approx \$4.77)
- Round 2: As receiver, sender sent \$0
- Round 3: I sent \$3 again
- Round 4: As receiver, sender sent \$0.68, I returned \$1.00

The earnings from Round 3 to Round 4:

$\$9.06 - \$10.10 =$ gained \$1.04 from Round 4 (which makes sense: \$2.04 received - \$1.00 returned)

So between Round 3 and Round 4:

Round 3 starting: unknown

Round 3 ending: \$9.06

Round 4 ending: \$10.10

From Round 3 (\$9.06) to Round 4 (\$10.10), I gained \$1.04, which matches being a receiver who got \$2.04 and returned \$1.00.

Let me recalculate from the beginning:

- Round 1: Sent \$3
 - Started with unknown
 - Kept \$2 from initial
 - Got back \$4.77 from receiver
 - Ended with \$6.77
- Round 2: Was receiver, sender sent \$0
 - Started with \$6.77
 - Received \$0, returned \$0
 - Ended with \$6.77
- Round 3: Sent \$3
 - Started with \$6.77
 - Kept \$2, sent \$3
 - Got back \$7.06 from receiver
 - Ended with \$9.06...

Usage: {

"input_tokens": 784,

"output_tokens": 854,

"reasoning_tokens": 0

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.390000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0. This means I have nothing to return - the maximum I can return is \$0.

This is interesting - in round 2, I sent \$3.50 (70% of my endowment) showing trust, but the sender this round sent me nothing. This could be:

1. A different sender in this round (multi-player game)
2. Communication noise caused them to receive/perceive something different
3. They chose not to send anything

Since I received \$0, I can only return \$0.

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 715,
  "output_tokens": 171,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.390000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

- Round 2: I sent \$3.50 and received \$0 back
- Round 3: I received \$0 from the sender

Cooperation seems to have broken down, possibly due to communication noise mentioned in the instructions. However, sending \$0 perpetuates the cycle of defection.

I'll send a moderate amount to signal willingness to cooperate while being somewhat more cautious:

```
{'send': 2.50}
```

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the RECEIVER. The sender sent me some amount (I received it multiplied by 3 x), and I returned some amount. My earnings suggest I kept some and returned some.

Round 2: I was the SENDER. I sent \$3.50 to the receiver. This was multiplied by 3x to become \$10.50 for the receiver. My total earnings after Round 2 were \$7.39.

- Started Round 2 with: \$5.89 (from Round 1)
- Received in Round 2: \$5 (endowment)
- Sent: \$3.50
- So I had: $\$5 - \$3.50 = \$1.50$ left
- Then I received back some amount from the receiver
- Total after Round 2 = Previous + \$1.50 + returned = \$5.89 + \$1.50 + returned = \$7.39
- So returned = $\$7.39 - \$5.89 = \$1.50$

Wait, let me recalculate:

- Before Round 2, I had: \$5.89
- In Round 2, I got \$5 endowment, sent \$3.50, so kept \$1.50
- Then receiver returned something to me
- After Round 2, total = \$7.39
- So: \$5.89 + \$1.50 + returned = \$7.39
- returned = \$7.39 - \$7.39 = \$0

Hmm, that means the receiver in Round 2 returned \$0 to me! I sent them \$3.50 (which became \$10.50), and they kept it all and returned nothing.

Round 3: I was the RECEIVER. The sender sent me \$0, so I received \$0 and returned \$0. My total stayed at \$7.39.
Now it's Round 4 and I'm sending again. Looking back, the receiver didn't return anything after I sent \$3.50 in Round 2, and then the sender sent me nothing in Round 3. There's communication noise that could distort amounts, so it's possible the receiver did send something back but it appeared as \$0, or the sender in Round 3 actually sent something that got lost. I need to decide whether to trust again or protect myself given this history of apparent defection.

I could send nothing to play it safe, try a moderate amount to signal willingness to cooperate while limiting risk, or send high again to demonstrate trust and potentially reset the dynamic. Since noise might be distorting what I'm seeing and cooperation seems worth attempting, I'll go with a middle ground-sending \$2.50 shows I'm still willing to engage without fully exposing myself to another potential loss....

usage:

```
{
  "input_tokens": 591,
  "output_tokens": 1069,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.870000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. I have nothing to return.

This is straightforward - if I received \$0, I can only return between 0 and 0, which is just 0

Usage: {

```
"input_tokens": 642,
"output_tokens": 86,
"reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0;

noise/visible: sent visible=\$0.51, received visible=\$1.53

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.75},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 28.0,

"Agent_2": 32.0

}

Visible balances after round: {

"Agent_1": 14.399999999999997,

"Agent_2": 16.87

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

- Round 3: I received \$0 from the sender

Cooperation seems to have broken down, possibly due to communication noise mentioned in the instructions. However, sending \$0 perpetuates the cycle of defection.

I'll send a moderate amount to signal willingness to cooperate while being somewhat more cautious:

```
{'send': 2.50}
```

reasoning:

Let me think about what happened in the previous rounds:

Round 1: I was the RECEIVER. The sender sent me some amount (I received it multiplied by 3x), and I returned some amount. My earnings suggest I kept some and returned some.

Round 2: I was the SENDER. I sent \$3.50 to the receiver. This was multiplied by 3x to become \$10.50 for the receiver. My total earnings after Round 2 were \$7.39.

- Started Round 2 with: \$5.89 (from Round 1)
- Received in Round 2: \$5 (endowment)
- Sent: \$3.50
- So I had: $\$5 - \$3.50 = \$1.50$ left
- Then I received back some amount from the receiver
- Total after Round 2 = Previous + \$1.50 + returned = $\$5.89 + \$1.50 + \text{returned} = \7.39
- So returned = $\$7.39 - \$7.39 = \$0$

Wait, let me recalculate:

- Before Round 2, I had: \$5.89
- In Round 2, I got \$5 endowment, sent \$3.50, so kept \$1.50
- Then receiver returned something to me
- After Round 2, total = \$7.39
- So: $\$5.89 + \$1.50 + \text{returned} = \7.39
- returned = $\$7.39 - \$7.39 = \$0$

Hmm, that means the receiver in Round 2 returned \$0 to me! I sent them \$3.50 (which became \$10.50), and they kept it all and returned nothing.

Round 3: I was the RECEIVER. The sender sent me \$0, so I received \$0 and returned \$0. My total stayed at \$7.39.

Now it's Round 4 and I'm sending again. Looking back, the receiver didn't return anything after I sent \$3.50 in Round 2, and then the sender sent me nothing in Round 3. There's communication noise that could distort amounts, so it's possible the receiver did send something back but it appeared as \$0, or the sender in Round 3 actually sent something that got lost. I need to decide whether to trust again or protect myself given this history of apparent defection.

I could send nothing to play it safe, try a moderate amount to signal willingness to cooperate while limiting risk, or send high again to demonstrate trust and potentially reset the dynamic. Since noise might be distorting what I'm seeing and cooperation seems worth attempting, I'll go with a middle ground-sending \$2.50 shows I'm still willing to engage without fully exposing myself to another potential loss....

usage:

```
{
  "input_tokens": 591,
  "output_tokens": 1069,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.870000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. I have nothing to return.

This is straightforward - if I received \$0, I can only return between 0 and 0, which is just 0.

usage:

```
{
  "input_tokens": 642,
  "output_tokens": 86,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.870000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- Round 2: I sent \$3.50, received \$0 back
- Round 4: I sent \$2.50, received \$0 back
- Round 3 & 5: When the other player was sender, they sent \$0

There's been consistent non-cooperation despite my attempts to signal willingness to cooperate. Continuing to send money when receiving nothing in return is not strategically sound.

```
{'send': 0}
```

Reasoning:

Let me analyze the pattern of play so far:

Round 2: I was SENDER - sent \$3.50, received \$0 back

Round 3: I was RECEIVER - received \$0 (sender sent \$0)

Round 4: I was SENDER - sent \$2.50, and based on my earnings staying at \$11.87, I must have received \$0 back

Round 5: I was RECEIVER - sender sent \$0, I returned \$0

The pattern shows complete breakdown of cooperation. Despite my attempts to cooperate by sending \$3.50 in round 2 and \$2.50 in round 4, I received nothing back. The other player has also been sending \$0 when they're the sender.

Given this pattern of non-cooperation and the mentioned communication noise, continuing to send money doesn't make strategic sense. The other player is consistently not reciprocating.

I should send \$0 to protect my earnings.

Usage: {

```
"input_tokens": 628,
"output_tokens": 326,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.059999999999999.

This round, The sender sent you \$0.68, so you received \$2.04. How much do you return to The sender? (0-2.04)

Message 3 role=assistant

content:

As RECEIVER this round:

- The sender sent \$0.68
- I received \$2.04 (after 3x multiplier)
- The sender showed some cooperation by sending a positive amount

To maintain reciprocity and encourage continued cooperation, I'll return a fair portion. Since the sender did cooperate (even if with a smaller amount), I should reward that trust.

```
{'return': 1.0}
```

This gives the sender a positive return on their investment (\$5 - \$0.68 + \$1.00 = \$5.32) while I keep \$1.04 from this round.

reasoning:

Let me analyze this situation:

As RECEIVER this round:

- The sender sent \$0.68
- I received $\$0.68 \times 3 = \2.04
- I need to decide how much to return (0-2.04)

My total earnings so far are \$9.06.

Looking at the pattern:

- Round 1: I was sender, sent \$3, got a good return (around \$4.77)
- Round 2: I was receiver, received \$0, returned \$0
- Round 3: I was sender, sent \$3
- Round 4: I am receiver, received \$2.04

The sender in Round 4 sent \$0.68, which is about 13.6% of their \$5 endowment. This is a relatively small amount compared to my \$3 sends, but it's still cooperative (non-zero).

Given that:

1. The note says there may be communication noise
2. The sender did cooperate by sending something
3. I've been cooperative in the past
4. Fair reciprocity would be to return a reasonable amount

A fair return might be:

- If I return about half (\$1.02), the sender would get: $\$5 - \$0.68 + \$1.02 = \5.34
- I would keep: $\$2.04 - \$1.02 = \$1.02$

Or I could be more generous:

- If I return \$1.36 (2/3), the sender would get: $\$5 - \$0.68 + \$1.36 = \5.68
- I would keep: $\$2.04 - \$1.36 = \$0.68$

My strategy has been cooperative. Given the sender sent a smaller amount, I should still return a fair portion. Returning about half to 2/3 seems reasonable. Let me return \$1.00 which is roughly half.

Actually, let me think about this more carefully. The sender sent \$0.68. After 3x, I got \$2.04.

...

usage:

```
{
  "input_tokens": 711,
  "output_tokens": 692,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.099999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my trajectory:

- Round 3: I sent \$3 (cooperative approach)
- Round 4: As receiver, sender sent \$0.68 (some cooperation), I returned \$1.00 to reciprocate

Total earnings increased from \$9.06 to \$10.10, showing positive outcomes from cooperation.

The pattern shows that cooperative play has been generally rewarding, though there's variation in partners' cooperation levels.

For Round 5, I'll maintain a cooperative strategy:

{'send': 3}

This amount:

- Creates value through the 3x multiplier (\$9 for receiver)
- Shows continued trustworthiness
- Has historically led to positive returns
- Balances risk with potential reward

reasoning:

Let me track my earnings history:

Round 1: I sent \$3, total earnings were \$6.77

Round 2: I was receiver, sender sent me \$0

Round 3: I sent \$3, total increased to \$9.06

Round 4: I was receiver, sender sent \$0.68, I received \$2.04, I returned \$1.00, total is now \$10.10

Let me verify Round 4:

- Started with \$9.06
- Received \$2.04
- Returned \$1.00
- New total: $\$9.06 + \$2.04 - \$1.00 = \10.10 ?

Now looking at the pattern:

- Round 1: I sent \$3, got back good return (approx \$4.77)
- Round 2: As receiver, sender sent \$0
- Round 3: I sent \$3 again
- Round 4: As receiver, sender sent \$0.68, I returned \$1.00

The earnings from Round 3 to Round 4:

$\$9.06 - \$10.10 =$ gained \$1.04 from Round 4 (which makes sense: \$2.04 received - \$1.00 returned)

So between Round 3 and Round 4:

Round 3 starting: unknown

Round 3 ending: \$9.06

Round 4 ending: \$10.10

From Round 3 (\$9.06) to Round 4 (\$10.10), I gained \$1.04, which matches being a receiver who got \$2.04 and returned \$1.00.

Let me recalculate from the beginning:

- Round 1: Sent \$3
 - Started with unknown
 - Kept \$2 from initial
 - Got back \$4.77 from receiver
 - Ended with \$6.77
- Round 2: Was receiver, sender sent \$0
 - Started with \$6.77
 - Received \$0, returned \$0
 - Ended with \$6.77
- Round 3: Sent \$3
 - Started with \$6.77
 - Kept \$2, sent \$3
 - Got back \$7.06 from receiver
 - Ended with \$9.06...

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.869999999999997.

This round, The sender sent you \$0.51, so you received \$1.53. How much do you return to The sender? (0-1.53)

Assistant output:

As RECEIVER this round:

- The sender sent \$0.51
- I received \$1.53 (after 3x multiplier)
- This is a relatively small amount, but still shows some trust


```

    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.099999999999999.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my trajectory:

- Round 3: I sent \$3 (cooperative approach)
- Round 4: As receiver, sender sent \$0.68 (some cooperation), I returned \$1.00 to reciprocate

Total earnings increased from \$9.06 to \$10.10, showing positive outcomes from cooperation.

The pattern shows that cooperative play has been generally rewarding, though there's variation in partners' cooperation levels.

For Round 5, I'll maintain a cooperative strategy:

{'send': 3}

This amount:

- Creates value through the 3x multiplier (\$9 for receiver)
- Shows continued trustworthiness
- Has historically led to positive returns
- Balances risk with potential reward

reasoning:

Let me track my earnings history:

Round 1: I sent \$3, total earnings were \$6.77

Round 2: I was receiver, sender sent me \$0

Round 3: I sent \$3, total increased to \$9.06

Round 4: I was receiver, sender sent \$0.68, I received \$2.04, I returned \$1.00, total is now \$10.10

Let me verify Round 4:

- Started with \$9.06
- Received \$2.04
- Returned \$1.00
- New total: $\$9.06 + \$2.04 - \$1.00 = \10.10 ?

Now looking at the pattern:

- Round 1: I sent \$3, got back good return (approx \$4.77)
- Round 2: As receiver, sender sent \$0
- Round 3: I sent \$3 again
- Round 4: As receiver, sender sent \$0.68, I returned \$1.00

The earnings from Round 3 to Round 4:

$\$9.06 - \$10.10 =$ gained \$1.04 from Round 4 (which makes sense: \$2.04 received - \$1.00 returned)

So between Round 3 and Round 4:

Round 3 starting: unknown

Round 3 ending: \$9.06

Round 4 ending: \$10.10

From Round 3 (\$9.06) to Round 4 (\$10.10), I gained \$1.04, which matches being a receiver who got \$2.04 and returned \$1.00.

Let me recalculate from the beginning:

- Round 1: Sent \$3
 - Started with unknown
 - Kept \$2 from initial
 - Got back \$4.77 from receiver
 - Ended with \$6.77
- Round 2: Was receiver, sender sent \$0
 - Started with \$6.77
 - Received \$0, returned \$0
 - Ended with \$6.77
- Round 3: Sent \$3
 - Started with \$6.77
 - Kept \$2, sent \$3
 - Got back \$7.06 from receiver
 - Ended with \$9.06...

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.869999999999997.

This round, The sender sent you \$0.51, so you received \$1.53. How much do you return to The sender? (0-1.53)

Message 5 role=assistant

content:

As RECEIVER this round:

- The sender sent \$0.51
- I received \$1.53 (after 3x multiplier)
- This is a relatively small amount, but still shows some trust

I'll return a proportional amount to maintain reciprocity while acknowledging the lower cooperation level:

{'return': 0.75}

This gives the sender a modest positive return ($\$5 - \$0.51 + \$0.75 = \5.24) and leaves me with \$0.78 from this round. This approach encourages continued cooperation without over-rewarding the lower send amount.

reasoning:

As RECEIVER this round:

- The sender sent \$0.51
- I received \$1.53 (after 3x multiplier)
- My total earnings are \$12.87

The sender sent a relatively small amount (\$0.51). This is lower cooperation than previous rounds I've seen. However, they did send something positive rather than \$0, so there's still some trust being shown.

I need to decide how much to return. Let me think about fairness and reciprocity:

- If I return too little, I discourage future cooperation
- If I return too much, I might be overly generous given the low initial send

Since they sent \$0.51 and it became \$1.53, a fair return might be around half or slightly less.

Let me return \$0.75. This would:

- Give the sender: $\$5 - \$0.51 + \$0.75 = \5.24 (positive return)
- Leave me with: $\$1.53 - \$0.75 = \$0.78$

This maintains some reciprocity while not being overly generous given the relatively low send amount.

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 400,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.399999999999997.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my recent history:

- Round 5: I sent \$3, my earnings increased by \$2.77 (suggesting I received about \$0.77 back from \$9 sent to receiver - low reciprocity)
- Round 6: As receiver, sender sent \$0.51, I returned \$0.75 (fairly generous)

The low return in Round 5 suggests I should adjust my strategy. I'll send a moderate amount that balances cooperation with risk management:

{'send': 2}

This creates \$6 for the receiver through the multiplier while protecting against very low return rates. It maintains some cooperation but adjusts for recent experience.

Reasoning:

Let me track my progress:

- Round 3: I sent \$3 as SENDER
- Round 4: As RECEIVER, sender sent \$0.68, I returned \$1.00
- Round 5: I sent \$3 as SENDER (my earnings went from \$10.10 to \$12.87)
- Round 6: As RECEIVER, sender sent \$0.51, I returned \$0.75 (my earnings went from \$12.87 to \$14.40)

Let me calculate Round 5's outcome:

- I sent \$3
- Receiver got \$9
- My earnings increased from \$10.10 to \$12.87 = \$2.77 gain
- Starting with \$5, sent \$3, so I had \$2
- To get \$2.77 total, I must have received \$0.77 back
- So the receiver returned \$0.77 out of \$9 (about 8.6%)

That's a pretty low return rate. The receiver only gave back about 8.6% of what they received.

Round 6 was clearer:

- Sender sent \$0.51, I received \$1.53
- I returned \$0.75
- My earnings went from \$12.87 to \$14.40 = \$1.53 gain
- Wait, that means I kept all \$1.53 and my calculation above must be wrong about the return

Let me recalculate Round 6:

- Starting earnings: \$12.87
- Ending earnings: \$14.40
- Difference: \$1.53
- I received \$1.53 this round
- I said I'd return \$0.75
- If I returned \$0.75, I'd keep \$1.53 - \$0.75 = \$0.78
- But my earnings only went up by \$1.53, which suggests I kept everything?

Actually, the total earnings shown is cumulative. So:

- If I received \$1.53 and returned \$0.75, I should net \$0.78
- But the earnings jumped by \$1.53, which doesn't align

There's a discrepancy here. Either the return wasn't processed yet, or there's something off with how the amounts are being recorded. Let me revisit Round 5 to see if there's a pattern-I sent \$3 and gained \$2.77 total, meaning I kept \$2 and received \$0.77 back from the receiver

Now for Round 7, I'm sending again. My track record shows I sent \$3 in Round 5 and got minimal reciprocity (\$0.77 back), while as a receiver I returned \$0.75 on \$1.53 (nearly 50%). The symmetry suggests that being generous as a sender doesn't guarantee returns, but I've been fairly reciprocal when receiving. I should probably send less this round given the poor return last time, though I need to account for different players and potential communication issues. A moderate amount seems like the right balance....

```
Usage: {
  "input_tokens": 802,
  "output_tokens": 1045,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    }
  }
}
```

```

    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.870000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0. I have nothing to return.

This is straightforward - if I received \$0, I can only return between 0 and 0, which is just 0.

usage:

```

{
  "input_tokens": 642,
  "output_tokens": 86,
  "reasoning_tokens": 0
}

```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.87000000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

- Round 2: I sent \$3.50, received \$0 back
- Round 4: I sent \$2.50, received \$0 back
- Round 3 & 5: When the other player was sender, they sent \$0

There's been consistent non-cooperation despite my attempts to signal willingness to cooperate. Continuing to send money when receiving nothing in return is not strategically sound.

{'send': 0}

reasoning:

Let me analyze the pattern of play so far:

Round 2: I was SENDER - sent \$3.50, received \$0 back

Round 3: I was RECEIVER - received \$0 (sender sent \$0)

Round 4: I was SENDER - sent \$2.50, and based on my earnings staying at \$11.87, I must have received \$0 back

Round 5: I was RECEIVER - sender sent \$0, I returned \$0

The pattern shows complete breakdown of cooperation. Despite my attempts to cooperate by sending \$3.50 in round 2 and \$2.50 in round 4, I received nothing back. The other player has also been sending \$0 when they're the sender.

Given this pattern of non-cooperation and the mentioned communication noise, continuing to send money doesn't make strategic sense. The other player is consistently not reciprocating.

I should send \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 628,
  "output_tokens": 326,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.87.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 and have nothing to return.

{'return': 0}

Reasoning:

I'm the receiver in round 7. The sender sent me \$0, so I received \$0 (which would be multiplied by 3, but $0 \times 3 = 0$).

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

{'return': 0}

Usage: {

```
"input_tokens": 640,
"output_tokens": 112,
"reasoning_tokens": 0
```

}

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0;

noise/visible: sent visible=\$0.43, received visible=\$1.29

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.5},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 31.0,

"Agent_2": 43.0

}

Visible balances after round: {

"Agent_1": 18.719999999999995,

"Agent_2": 21.87

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$11.870000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

- Round 2: I sent \$3.50, received \$0 back
- Round 4: I sent \$2.50, received \$0 back
- Round 3 & 5: When the other player was sender, they sent \$0

There's been consistent non-cooperation despite my attempts to signal willingness to cooperate rate. Continuing to send money when receiving nothing in return is not strategically sound.

{'send': 0}

reasoning:

Let me analyze the pattern of play so far:

Round 2: I was SENDER - sent \$3.50, received \$0 back

Round 3: I was RECEIVER - received \$0 (sender sent \$0)

Round 4: I was SENDER - sent \$2.50, and based on my earnings staying at \$11.87, I must have received \$0 back

Round 5: I was RECEIVER - sender sent \$0, I returned \$0

The pattern shows complete breakdown of cooperation. Despite my attempts to cooperate by sending \$3.50 in round 2 and \$2.50 in round 4, I received nothing back. The other player has also been sending \$0 when they're the sender.

Given this pattern of non-cooperation and the mentioned communication noise, continuing to send money doesn't make strategic sense. The other player is consistently not reciprocating.

I should send \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 628,
  "output_tokens": 326,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.87.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

{'return': 0}

reasoning:

I'm the receiver in round 7. The sender sent me \$0, so I received \$0 (which would be multi

plied by 3, but $0 \times 3 = 0$).

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

```
{'return': 0}
usage:
{
  "input_tokens": 640,
  "output_tokens": 112,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.87.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete pattern:

- When I sent money (Rounds 2, 4): received \$0 back
- When the other player was sender (Rounds 3, 5, 7): they sent \$0
- Round 6: I sent \$0 after recognizing the pattern

There has been zero cooperation or reciprocity throughout this game. Sending money would only decrease my earnings with no expected return.

```
{'send': 0}
```

Reasoning:

Let me review the pattern:

- Round 2: I sent \$3.50, received \$0 back
- Round 4: I sent \$2.50, received \$0 back
- Round 6: I sent \$0 (after learning my lesson)
- Round 7: I was receiver, sender sent me \$0, I returned \$0

The pattern is clear - there's been no cooperation or reciprocity in this game. The other player:

1. Sends \$0 when they're the sender
2. Returns \$0 when they're the receiver

I've tried cooperating early on but got nothing in return. Sending money at this point would just reduce my earnings with no expected return.

The rational decision is to send \$0.

Usage: {

```
"input_tokens": 622,
"output_tokens": 278,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
```



```

"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.869999999999997.

This round, The sender sent you \$0.51, so you received \$1.53. How much do you return to The sender? (0-1.53)

Message 3 role=assistant

content:

As RECEIVER this round:

- The sender sent \$0.51
- I received \$1.53 (after 3x multiplier)
- This is a relatively small amount, but still shows some trust

I'll return a proportional amount to maintain reciprocity while acknowledging the lower cooperation level:

```
{'return': 0.75}
```

This gives the sender a modest positive return ($\$5 - \$0.51 + \$0.75 = \5.24) and leaves me with \$0.78 from this round. This approach encourages continued cooperation without over-rewarding the lower send amount.

reasoning:

As RECEIVER this round:

- The sender sent \$0.51
- I received \$1.53 (after 3x multiplier)

- My total earnings are \$12.87

The sender sent a relatively small amount (\$0.51). This is lower cooperation than previous rounds I've seen. However, they did send something positive rather than \$0, so there's still some trust being shown.

I need to decide how much to return. Let me think about fairness and reciprocity:

- If I return too little, I discourage future cooperation
- If I return too much, I might be overly generous given the low initial send

Since they sent \$0.51 and it became \$1.53, a fair return might be around half or slightly less.

Let me return \$0.75. This would:

- Give the sender: $\$5 - \$0.51 + \$0.75 = \5.24 (positive return)
- Leave me with: $\$1.53 - \$0.75 = \$0.78$

This maintains some reciprocity while not being overly generous given the relatively low send amount.

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 400,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.399999999999999.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my recent history:

- Round 5: I sent \$3, my earnings increased by \$2.77 (suggesting I received about \$0.77 back from \$9 sent to receiver - low reciprocity)
- Round 6: As receiver, sender sent \$0.51, I returned \$0.75 (fairly generous)

The low return in Round 5 suggests I should adjust my strategy. I'll send a moderate amount that balances cooperation with risk management:

```
{'send': 2}
```

This creates \$6 for the receiver through the multiplier while protecting against very low return rates. It maintains some cooperation but adjusts for recent experience.

reasoning:

Let me track my progress:

- Round 3: I sent \$3 as SENDER
- Round 4: As RECEIVER, sender sent \$0.68, I returned \$1.00
- Round 5: I sent \$3 as SENDER (my earnings went from \$10.10 to \$12.87)
- Round 6: As RECEIVER, sender sent \$0.51, I returned \$0.75 (my earnings went from \$12.87 to \$14.40)

Let me calculate Round 5's outcome:

- I sent \$3
- Receiver got \$9
- My earnings increased from \$10.10 to \$12.87 = \$2.77 gain
- Starting with \$5, sent \$3, so I had \$2
- To get \$2.77 total, I must have received \$0.77 back
- So the receiver returned \$0.77 out of \$9 (about 8.6%)

That's a pretty low return rate. The receiver only gave back about 8.6% of what they received.

Round 6 was clearer:

- Sender sent \$0.51, I received \$1.53
- I returned \$0.75
- My earnings went from \$12.87 to \$14.40 = \$1.53 gain
- Wait, that means I kept all \$1.53 and my calculation above must be wrong about the return

n

Let me recalculate Round 6:

- Starting earnings: \$12.87
- Ending earnings: \$14.40
- Difference: \$1.53
- I received \$1.53 this round
- I said I'd return \$0.75
- If I returned \$0.75, I'd keep $\$1.53 - \$0.75 = \$0.78$
- But my earnings only went up by \$1.53, which suggests I kept everything?

Actually, the total earnings shown is cumulative. So:

- If I received \$1.53 and returned \$0.75, I should net \$0.78
- But the earnings jumped by \$1.53, which doesn't align

There's a discrepancy here. Either the return wasn't processed yet, or there's something off with how the amounts are being recorded. Let me revisit Round 5 to see if there's a pattern-I sent \$3 and gained \$2.77 total, meaning I kept \$2 and received \$0.77 back from the receiver.

Now for Round 7, I'm sending again. My track record shows I sent \$3 in Round 5 and got minimal reciprocity (\$0.77 back), while as a receiver I returned \$0.75 on \$1.53 (nearly 50%). The asymmetry suggests that being generous as a sender doesn't guarantee returns, but I've been fairly reciprocal when receiving. I should probably send less this round given the poor return last time, though I need to account for different players and potential communication issues. A moderate amount seems like the right balance....

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 1045,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.429999999999996.

This round, The sender sent you \$0.43, so you received \$1.29. How much do you return to The sender? (0-1.29)

Assistant output:

As RECEIVER this round:

- The sender sent \$0.43
- I received \$1.29 (after 3x multiplier)
- This is even lower than Round 6's \$0.51

The declining send amounts suggest decreasing trust or cooperation. I'll return a proportional amount, but adjusted downward to reflect the lower initial send:

```
{'return': 0.50}
```

This gives the sender a small positive return ($\$5 - \$0.43 + \$0.50 = \5.07) and leaves me with \$0.79. This maintains some reciprocity while signaling that lower cooperation levels receive proportionally lower returns.

Reasoning:

As RECEIVER this round:

- The sender sent \$0.43
- I received \$1.29 (after 3x multiplier)
- This is even lower than round 6 when they sent \$0.51

Looking at the pattern:

- Round 6: Sender sent me \$0.51, I returned \$0.75
- Round 7: I was sender and sent \$2
- Round 8: Now sender sent me only \$0.43

This is a declining cooperation pattern. The sender is sending less and less. However, I should maintain some reciprocity to encourage future cooperation. But I should also adjust downward given the lower amounts being sent.

In Round 6, sender sent \$0.51 and I returned \$0.75 (about 49% of what I received).

This round they sent even less (\$0.43, receiving \$1.29).

I could:

1. Return nothing - punish low cooperation
2. Return a small amount like \$0.50-0.60 - maintain some reciprocity but adjusted for lower trust
3. Return proportionally similar to last time

Given the declining cooperation, I'll return a bit less proportionally than before, maybe around 40% of what I received:

$\$1.29 * 0.4 = \0.52

This keeps some cooperation alive but signals that lower sends get lower returns.

```
Usage: {
  "input_tokens": 807,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$1; received \$3; Agent_2 returned \$0.45; payoffs Agent_1=\$4.45, Agent_2=\$2.55; noise/visible: sent visible=\$0.31, received visible=\$0.93, returned visible=\$0.41
Action validation: {"returned": {"amount": 0.45, "clamped": false, "max": 3.0, "min": 0, "raw": 0.45}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 35.45,
"Agent_2": 45.55

}

Visible balances after round: {

"Agent_1": 23.129999999999995,
"Agent_2": 22.35

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.399999999999999.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my recent history:

- Round 5: I sent \$3, my earnings increased by \$2.77 (suggesting I received about \$0.77 back from \$9 sent to receiver - low reciprocity)
- Round 6: As receiver, sender sent \$0.51, I returned \$0.75 (fairly generous)

The low return in Round 5 suggests I should adjust my strategy. I'll send a moderate amount that balances cooperation with risk management:

{'send': 2}

This creates \$6 for the receiver through the multiplier while protecting against very low return rates. It maintains some cooperation but adjusts for recent experience.

reasoning:

Let me track my progress:

- Round 3: I sent \$3 as SENDER
- Round 4: As RECEIVER, sender sent \$0.68, I returned \$1.00
- Round 5: I sent \$3 as SENDER (my earnings went from \$10.10 to \$12.87)
- Round 6: As RECEIVER, sender sent \$0.51, I returned \$0.75 (my earnings went from \$12.87 to \$14.40)

Let me calculate Round 5's outcome:

- I sent \$3
- Receiver got \$9
- My earnings increased from \$10.10 to \$12.87 = \$2.77 gain
- Starting with \$5, sent \$3, so I had \$2

- To get \$2.77 total, I must have received \$0.77 back
- So the receiver returned \$0.77 out of \$9 (about 8.6%)

That's a pretty low return rate. The receiver only gave back about 8.6% of what they received.

Round 6 was clearer:

- Sender sent \$0.51, I received \$1.53
- I returned \$0.75
- My earnings went from \$12.87 to \$14.40 = \$1.53 gain
- Wait, that means I kept all \$1.53 and my calculation above must be wrong about the return

Let me recalculate Round 6:

- Starting earnings: \$12.87
- Ending earnings: \$14.40
- Difference: \$1.53
- I received \$1.53 this round
- I said I'd return \$0.75
- If I returned \$0.75, I'd keep $\$1.53 - \$0.75 = \$0.78$
- But my earnings only went up by \$1.53, which suggests I kept everything?

Actually, the total earnings shown is cumulative. So:

- If I received \$1.53 and returned \$0.75, I should net \$0.78
- But the earnings jumped by \$1.53, which doesn't align

There's a discrepancy here. Either the return wasn't processed yet, or there's something off with how the amounts are being recorded. Let me revisit Round 5 to see if there's a pattern-I sent \$3 and gained \$2.77 total, meaning I kept \$2 and received \$0.77 back from the receiver.

Now for Round 7, I'm sending again. My track record shows I sent \$3 in Round 5 and got minimal reciprocity (\$0.77 back), while as a receiver I returned \$0.75 on \$1.53 (nearly 50%). The asymmetry suggests that being generous as a sender doesn't guarantee returns, but I've been fairly reciprocal when receiving. I should probably send less this round given the poor return last time, though I need to account for different players and potential communication issues. A moderate amount seems like the right balance....

usage:

```
{
  "input_tokens": 802,
  "output_tokens": 1045,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.429999999999996.

This round, The sender sent you \$0.43, so you received \$1.29. How much do you return to The sender? (0-1.29)

Message 5 role=assistant

content:

As RECEIVER this round:

- The sender sent \$0.43
- I received \$1.29 (after 3x multiplier)
- This is even lower than Round 6's \$0.51

The declining send amounts suggest decreasing trust or cooperation. I'll return a proportional amount, but adjusted downward to reflect the lower initial send:

```
{'return': 0.50}
```

This gives the sender a small positive return ($\$5 - \$0.43 + \$0.50 = \5.07) and leaves me with \$0.79. This maintains some reciprocity while signaling that lower cooperation levels receive proportionally lower returns.

reasoning:

As RECEIVER this round:

- The sender sent \$0.43
- I received \$1.29 (after 3x multiplier)

- This is even lower than round 6 when they sent \$0.51

Looking at the pattern:

- Round 6: Sender sent me \$0.51, I returned \$0.75
- Round 7: I was sender and sent \$2
- Round 8: Now sender sent me only \$0.43

This is a declining cooperation pattern. The sender is sending less and less. However, I should maintain some reciprocity to encourage future cooperation. But I should also adjust downward given the lower amounts being sent.

In Round 6, sender sent \$0.51 and I returned \$0.75 (about 49% of what I received). This round they sent even less (\$0.43, receiving \$1.29).

I could:

1. Return nothing - punish low cooperation
2. Return a small amount like \$0.50-0.60 - maintain some reciprocity but adjusted for lower trust
3. Return proportionally similar to last time

Given the declining cooperation, I'll return a bit less proportionally than before, maybe around 40% of what I received:

$$\$1.29 * 0.4 = \$0.52$$

This keeps some cooperation alive but signals that lower sends get lower returns.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$18.7199999999999995.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my recent experience:

Round 7: I sent \$2 and my earnings increased by about \$3.53, meaning I received approximately \$0.53 back from the \$6 sent to the receiver (return rate ~9%)

Round 8: As receiver, I got \$0.43 sent (received \$1.29), and I returned \$0.50

The consistent pattern of low returns when I send suggests I should reduce my sending to minimize losses. I'll send a smaller amount:

```
{'send': 1}
```

This creates \$3 for the receiver while limiting my exposure. If cooperation improves, I can adjust upward in future rounds.

Reasoning:

Let me track my recent history as sender:

Round 7: I sent \$2 (I should have received some return)

- My earnings were \$17.43 before Round 8
- My earnings are now \$18.72
- Increase: \$18.72 - \$17.43 = \$1.29

Wait, that doesn't match up. Let me recalculate:

- Round 7: I was sender, sent \$2
- Round 8: I was receiver, got \$1.29, returned \$0.50, so I kept \$0.79
- Total increase from Round 7 to now: \$18.72 - \$14.40 = \$4.32

That's not quite right either. Let me think about this more carefully:

Start of Round 7: \$14.40

Round 7 (SENDER): Sent \$2
- I keep: \$5 - \$2 = \$3
- Plus whatever was returned to me

Round 8 (RECEIVER): Got \$1.29, returned \$0.50
- I keep: \$1.29 - \$0.50 = \$0.79

So from Round 7 start (\$14.40) to Round 9 start (\$18.72):
Total gain = \$18.72 - \$14.40 = \$4.32

Round 7: \$3 (kept from initial) + return amount
Round 8: \$0.79

If Round 8 gave me \$0.79, then Round 7 must have given me:
\$4.32 - \$0.79 = \$3.53

So in Round 7, I sent \$2, kept \$3, and received back \$0.53 from the \$6 sent to the receiver.
That means receiver got \$6, returned \$0.53, kept \$5.47.
Return rate: \$0.53/\$6 = 8.8% - very low!

...

```
Usage: {  
  "input_tokens": 797,  
  "output_tokens": 749,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.87.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

{'return': 0}

reasoning:

I'm the receiver in round 7. The sender sent me \$0, so I received \$0 (which would be multiplied by 3, but $0 \times 3 = 0$).

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

{'return': 0}

usage:

```
{
  "input_tokens": 640,
  "output_tokens": 112,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.87.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the complete pattern:

- When I sent money (Rounds 2, 4): received \$0 back
- When the other player was sender (Rounds 3, 5, 7): they sent \$0
- Round 6: I sent \$0 after recognizing the pattern

There has been zero cooperation or reciprocity throughout this game. Sending money would only decrease my earnings with no expected return.

{'send': 0}

reasoning:

Let me review the pattern:

- Round 2: I sent \$3.50, received \$0 back
- Round 4: I sent \$2.50, received \$0 back
- Round 6: I sent \$0 (after learning my lesson)
- Round 7: I was receiver, sender sent me \$0, I returned \$0

The pattern is clear - there's been no cooperation or reciprocity in this game. The other

player:

1. Sends \$0 when they're the sender
2. Returns \$0 when they're the receiver

I've tried cooperating early on but got nothing in return. Sending money at this point would just reduce my earnings with no expected return.

The rational decision is to send \$0.

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 278,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$21.87.

This round, The sender sent you \$0.31, so you received \$0.9299999999999999. How much do you return to The sender? (0-0.9299999999999999)

Assistant output:

After 7 rounds of zero cooperation from both sides, the sender has made a small cooperative gesture by sending \$0.31. I'll reciprocate to encourage future cooperation.

They sent \$0.31 and I received \$0.93. I'll return approximately half of what I received to show goodwill while being cautious given the history.

```
{'return': 0.45}
```

Reasoning:

Interesting! The sender sent me \$0.31, which was multiplied by 3x to give me \$0.93.

This is the first time in the entire game that the sender has sent me anything. After 7 rounds of no cooperation, they're now sending a small amount.

Let me think about this:

1. They sent \$0.31 out of \$5
2. I received \$0.93
3. My total earnings are \$21.87

Given that:

- This is the first cooperative gesture I've seen
- They only sent a small amount (about 6% of their endowment)
- The note mentions there may be communication noise

I should consider:

- If I return nothing, I discourage future cooperation
- If I return something, I might encourage cooperation
- The amount is small, so this might be a test

A reasonable return might be to give them back roughly what they sent, or a bit more to show goodwill. They sent \$0.31 and kept \$4.69. If I return around \$0.31-\$0.50, they would end up with around \$5.00-\$5.19 total, and I'd keep \$0.43-\$0.62.

Actually, let me think about fairness. If we both cooperate fully:

- They send \$5, I receive \$15
- I return \$10, we both get \$10

In this case:

- They sent \$0.31 (6.2% of endowment)
- I received \$0.93
- A proportional return would be about 1/3 to 1/2 of what I received

I'll return about \$0.40-\$0.50 to show I'm willing to cooperate while keeping a reasonable amount since they only sent a small portion of their endowment.

Let me return \$0.45 - that gives them back more than they sent, showing goodwill.

Usage: {

```
"input_tokens": 644,
"output_tokens": 531,
```

```
"reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$0; payoffs Agent_2=\$3, Agent_1=\$6;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 41.45,

"Agent_2": 48.55

}

Visible balances after round: {

"Agent_1": 23.129999999999995,

"Agent_2": 25.35

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 10,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.87.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the complete pattern:

- When I sent money (Rounds 2, 4): received \$0 back
- When the other player was sender (Rounds 3, 5, 7): they sent \$0
- Round 6: I sent \$0 after recognizing the pattern

There has been zero cooperation or reciprocity throughout this game. Sending money would only decrease my earnings with no expected return.

{'send': 0}

reasoning:

Let me review the pattern:

- Round 2: I sent \$3.50, received \$0 back
- Round 4: I sent \$2.50, received \$0 back
- Round 6: I sent \$0 (after learning my lesson)
- Round 7: I was receiver, sender sent me \$0, I returned \$0

The pattern is clear - there's been no cooperation or reciprocity in this game. The other player:

1. Sends \$0 when they're the sender
2. Returns \$0 when they're the receiver

I've tried cooperating early on but got nothing in return. Sending money at this point would just reduce my earnings with no expected return.

The rational decision is to send \$0.

usage:

```
{
  "input_tokens": 622,
  "output_tokens": 278,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$21.87.

This round, The sender sent you \$0.31, so you received \$0.9299999999999999. How much do you return to The sender? (0-0.9299999999999999)

Message 5 role=assistant

content:

After 7 rounds of zero cooperation from both sides, the sender has made a small cooperative gesture by sending \$0.31. I'll reciprocate to encourage future cooperation.

They sent \$0.31 and I received \$0.93. I'll return approximately half of what I received to show goodwill while being cautious given the history.

```
{'return': 0.45}
```

reasoning:

Interesting! The sender sent me \$0.31, which was multiplied by 3x to give me \$0.93.

This is the first time in the entire game that the sender has sent me anything. After 7 rounds of no cooperation, they're now sending a small amount.

Let me think about this:

1. They sent \$0.31 out of \$5
2. I received \$0.93
3. My total earnings are \$21.87

Given that:

- This is the first cooperative gesture I've seen
- They only sent a small amount (about 6% of their endowment)
- The note mentions there may be communication noise

I should consider:

- If I return nothing, I discourage future cooperation
- If I return something, I might encourage cooperation
- The amount is small, so this might be a test

A reasonable return might be to give them back roughly what they sent, or a bit more to show goodwill. They sent \$0.31 and kept \$4.69. If I return around \$0.31-\$0.50, they would end up with around \$5.00-\$5.19 total, and I'd keep \$0.43-\$0.62.

Actually, let me think about fairness. If we both cooperate fully:

- They send \$5, I receive \$15
- I return \$10, we both get \$10

In this case:

- They sent \$0.31 (6.2% of endowment)
- I received \$0.93
- A proportional return would be about 1/3 to 1/2 of what I received

I'll return about \$0.40-\$0.50 to show I'm willing to cooperate while keeping a reasonable amount since they only sent a small portion of their endowment.

Let me return \$0.45 - that gives them back more than they sent, showing goodwill.

usage:

```
{
  "input_tokens": 644,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$22.35.

How much do you send to the receiver this round? (0-5)

Assistant output:

The receiver finally broke the pattern of zero cooperation in Round 9 by sending \$0.31, and I reciprocated by returning \$0.45. This is the first positive interaction in the entire game.

I should encourage this cooperative trend while still being cautious given the long history of non-cooperation. I'll send a moderate amount to test if we can build mutual trust.

```
{'send': 2}
```

Reasoning:

Let me analyze what happened in Round 9:

- The sender sent me \$0.31
- I received \$0.93
- I returned \$0.45

This was the first cooperative move in the entire game after 7 rounds of zero cooperation. I responded positively by returning \$0.45, which is roughly half of what I received.

Now in Round 10, I'm the sender again. The receiver showed willingness to cooperate by:

1. Making the first cooperative move after a long stalemate
2. Returning a reasonable amount (\$0.45 out of \$0.93)

I should reciprocate this cooperative gesture to build trust and potentially establish a mutually beneficial pattern. However, I need to be cautious given the history of zero cooperation for most of the game.

I'll send a moderate amount to test if cooperation will continue. Sending around \$2-3 seems reasonable - enough to show I'm willing to cooperate but not so much that I risk too much if they don't reciprocate.

Let me calculate potential outcomes:

- If I send \$2: they receive \$6, if they return \$3, I get $\$5 - 2 + 3 = \6 (profit of \$1)
- If I send \$3: they receive \$9, if they return \$4.5, I get $\$5 - 3 + 4.5 = \6.5 (profit of \$1.5)

Given that they showed cooperation but it's still early in rebuilding trust, I'll send \$2 as a measured cooperative response.

```
Usage: {  
  "input_tokens": 688,  
  "output_tokens": 438,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.429999999999996.

This round, The sender sent you \$0.43, so you received \$1.29. How much do you return to The sender? (0-1.29)

Message 3 role=assistant

content:

As RECEIVER this round:

- The sender sent \$0.43
- I received \$1.29 (after 3x multiplier)
- This is even lower than Round 6's \$0.51

The declining send amounts suggest decreasing trust or cooperation. I'll return a proportional amount, but adjusted downward to reflect the lower initial send:

{'return': 0.50}

This gives the sender a small positive return ($\$5 - \$0.43 + \$0.50 = \5.07) and leaves me with \$0.79. This maintains some reciprocity while signaling that lower cooperation levels receive proportionally lower returns.

reasoning:

As RECEIVER this round:

- The sender sent \$0.43
- I received \$1.29 (after 3x multiplier)
- This is even lower than round 6 when they sent \$0.51

Looking at the pattern:

- Round 6: Sender sent me \$0.51, I returned \$0.75
- Round 7: I was sender and sent \$2
- Round 8: Now sender sent me only \$0.43

This is a declining cooperation pattern. The sender is sending less and less. However, I should maintain some reciprocity to encourage future cooperation. But I should also adjust downward given the lower amounts being sent.

In Round 6, sender sent \$0.51 and I returned \$0.75 (about 49% of what I received).

This round they sent even less (\$0.43, receiving \$1.29).

I could:

1. Return nothing - punish low cooperation
2. Return a small amount like \$0.50-0.60 - maintain some reciprocity but adjusted for lower trust
3. Return proportionally similar to last time

Given the declining cooperation, I'll return a bit less proportionally than before, maybe around 40% of what I received:

$\$1.29 * 0.4 = \0.52

This keeps some cooperation alive but signals that lower sends get lower returns.

usage:

```
{  
  "input_tokens": 807,  
  "output_tokens": 465,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$18.71999999999999995.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my recent experience:

Round 7: I sent \$2 and my earnings increased by about \$3.53, meaning I received approximately \$0.53 back from the \$6 sent to the receiver (return rate ~9%)

Round 8: As receiver, I got \$0.43 sent (received \$1.29), and I returned \$0.50

The consistent pattern of low returns when I send suggests I should reduce my sending to minimize losses. I'll send a smaller amount:

```
{'send': 1}
```

This creates \$3 for the receiver while limiting my exposure. If cooperation improves, I can adjust upward in future rounds.

reasoning:

Let me track my recent history as sender:

Round 7: I sent \$2 (I should have received some return)

- My earnings were \$17.43 before Round 8
- My earnings are now \$18.72
- Increase: $\$18.72 - \$17.43 = \$1.29$

Wait, that doesn't match up. Let me recalculate:

- Round 7: I was sender, sent \$2
- Round 8: I was receiver, got \$1.29, returned \$0.50, so I kept \$0.79
- Total increase from Round 7 to now: $\$18.72 - \$14.40 = \$4.32$

That's not quite right either. Let me think about this more carefully:

Start of Round 7: \$14.40

Round 7 (SENDER): Sent \$2

- I keep: $\$5 - \$2 = \$3$
- Plus whatever was returned to me

Round 8 (RECEIVER): Got \$1.29, returned \$0.50

- I keep: $\$1.29 - \$0.50 = \$0.79$

So from Round 7 start (\$14.40) to Round 9 start (\$18.72):

Total gain = $\$18.72 - \$14.40 = \$4.32$

Round 7: \$3 (kept from initial) + return amount

Round 8: \$0.79

If Round 8 gave me \$0.79, then Round 7 must have given me:

$\$4.32 - \$0.79 = \$3.53$

So in Round 7, I sent \$2, kept \$3, and received back \$0.53 from the \$6 sent to the receiver.

That means receiver got \$6, returned \$0.53, kept \$5.47.

Return rate: $\$0.53/\$6 = 8.8\%$ - very low!

...

usage:

```
{
```



```
"input_tokens": 797,  
"output_tokens": 749,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$23.129999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

As RECEIVER this round:

- The sender sent \$0
- I received \$0 (after 3x multiplier)
- There's nothing to return

```
{'return': 0}
```

The sender chose not to send anything this round, so there are no funds to return.

Reasoning:

The sender sent \$0, so I received \$0 (after 3x multiplier).

Since I received nothing, there's nothing to return. The return must be between 0 and 0, so the only valid response is:

```
{'return': 0}
```

This makes sense - if the sender sends nothing, there's nothing for me to return.

Usage: {

```
"input_tokens": 809,  
"output_tokens": 149,  
"reasoning_tokens": 0  
}
```